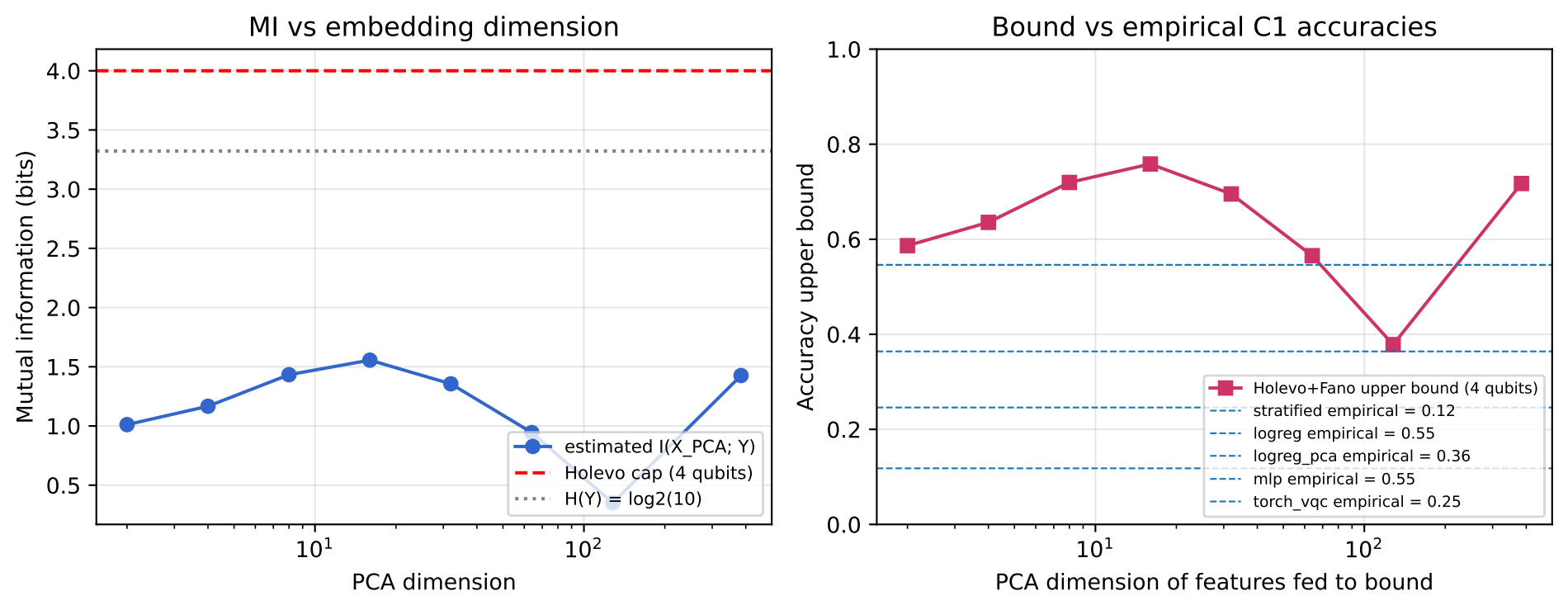
MI vs embedding dimension
Mutual information (bits)
PCA dimension
estimated I(X_PCA; Y)
Holevo cap (4 qubits)
H(Y) = log2(10)
Bound vs empirical C1 accuracies
Accuracy upper bound
PCA dimension of features fed to bound
Holevo+Fano upper bound (4 qubits)
stratified empirical = 0.12
logreg empirical = 0.55
logreg_pca empirical = 0.36
mlp empirical = 0.55
torch_vqc empirical = 0.25